

Backlog Buddy

Demonstration question paper generated from available previous papers - for practice only

B.Tech. I Semester Examination, 2026

Regulation: R20 | University: Andhra University

Digital Design (21EC32)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Number Systems & Boolean Algebra]

- a) Simplify the given Boolean function using a Karnaugh map.
- b) Explain the theorems of Boolean algebra with examples.

UNIT - 2 [Combinational Logic Design]

- a) Design a 4-bit binary adder using full adders.
- b) Implement the given Boolean function using an 8:1 multiplexer.

UNIT - 3 [Sequential Logic Design]

- a) Explain the working of SR, JK, D and T flip flops.
- b) Explain the working of SR, JK, D and T flip flops.

UNIT - 4 [Counters, Registers & State Machines]

- a) Explain the difference between asynchronous and synchronous counters.
- b) Design a mod-10 counter using flip flops.

UNIT - 5 [Logic Families, PLDs & FPGA]

- a) Compare TTL and CMOS logic families.
- b) Explain PLA, PAL and FPGA architectures.

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